REFONTE INTERNATIONAL TRAINING AND INTERNSHIP PROGRAM

Astrodynamics Specialist

SESSION 2

# Perturbations and Station-Keeping

*Why Real Orbits Drift, and How We Hold Them in Place*


**Prepared for** Cao Khanh Tan

**Module** 2, opening the Core Training Phase

**Builds on** Session 1: vis-viva, orbital elements, the GTO you sized

**Tools** poliastro / astropy (Cowell + J2), STK HPOP, GMAT force models

SESSION 2

# Perturbations and Station-Keeping

In Session 1 the only force was point-mass gravity, and the six orbital elements stayed fixed forever. Real orbits are not so tidy. The Earth is not a perfect sphere, the atmosphere still bites at low altitude, and the Sun and Moon pull on everything. These perturbations make the elements evolve with time. This module explains which forces matter, how the largest of them (Earth oblateness, the J2 effect) makes the orbit plane and the ellipse slowly rotate, and how mission designers turn that drift into a feature (sun-synchronous and Molniya orbits) or fight it with station-keeping.

## Learning objectives

- Write the perturbed equation of motion and list the main perturbing accelerations in order of size.
- Explain what J2 does to a, e, i, RAAN and argument of periapsis on average.
- Use the secular J2 rates to design a sun-synchronous orbit and to find the critical inclination.
- Describe how drag, third-body and solar radiation pressure act, and where each one dominates.
- Estimate a station-keeping delta-v budget and connect it to the vis-viva tools from Session 1.
- Propagate an orbit with J2 in code and confirm the secular drift matches the analytic formula.

> **Osculating vs mean elements**
> At any instant you can still describe a perturbed orbit with six elements: these are the osculating elements, the ellipse the satellite would follow if all perturbations switched off right now. They wobble every revolution. The mean elements average that wobble away and reveal the slow, secular trends we design around. When we say "J2 makes RAAN drift", we mean the mean RAAN.

PART 1

# The Perturbation Budget

We keep the two-body equation and add a perturbing acceleration on the right:

$$\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_p$$

The perturbing term is the sum of every non-ideal force per unit mass. Integrating this full expression numerically is called Cowell method, the workhorse of modern propagators:

$$\mathbf{a}_p = \mathbf{a}_{J_2} + \mathbf{a}_{\text{drag}} + \mathbf{a}_{3-\text{body}} + \mathbf{a}_{\text{SRP}} + \cdots$$

Not all terms matter everywhere. Their relative size depends almost entirely on altitude, as Figure 1 shows.

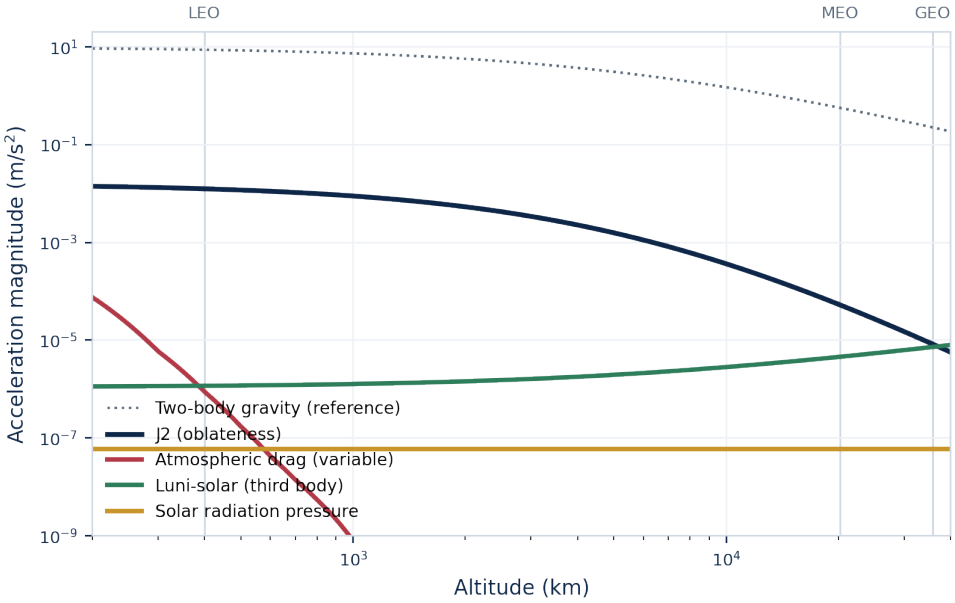



*Figure 1. Order-of-magnitude of the main perturbations against altitude (circular orbits). J2 dominates across most of LEO and MEO. Atmospheric drag only overtakes the others at low LEO, and it swings by orders of magnitude with solar activity and the ballistic coefficient (hence the single curve is only indicative). Luni-solar effects grow with altitude and rival J2 near GEO, where solar radiation pressure also becomes significant.*

PART 2

## The J2 Effect: Earth Oblateness

The Earth is an oblate spheroid: it bulges at the equator by about 21 km. That extra equatorial mass is described by the second zonal harmonic J2 = 1.0826 x 10^-3, and it is by far the largest perturbation for most satellites. Its acceleration in Earth-centred inertial coordinates is:

$$\mathbf{a}_{J_2} = -\frac{3J_2\mu R_E^2}{2r^4}\left[\left(1-5\frac{z^2}{r^2}\right)\frac{x}{r}\,\hat{\mathbf{x}} + \left(1-5\frac{z^2}{r^2}\right)\frac{y}{r}\,\hat{\mathbf{y}} + \left(3-5\frac{z^2}{r^2}\right)\frac{z}{r}\,\hat{\mathbf{z}}\right]$$

The important physics is what J2 does on average. Averaged over one revolution, J2 does not change the size (a), shape (e) or tilt (i) of the orbit. Instead it makes two of the orientation angles rotate steadily:

- **Nodal regression.** The orbital plane swivels about the Earth polar axis, so RAAN drifts.
- **Apsidal rotation.** The ellipse rotates within its plane, so the argument of periapsis drifts.

The secular rates are:

$$\dot{\Omega} = -\frac{3}{2}J_2\, n\left(\frac{R_E}{p}\right)^2 \cos i$$

$$\dot{\omega} = \frac{3}{4}J_2\, n\left(\frac{R_E}{p}\right)^2 (5\cos^2 i - 1)$$

with the mean motion and semi-latus rectum from Session 1:

$$n = \sqrt{\frac{\mu}{a^3}}, \qquad p = a(1-e^2)$$

**Read the signs**

For a prograde orbit (i < 90 deg), cos i > 0, so RAAN drift is negative: the node moves westward. For a retrograde orbit (i > 90 deg) it moves eastward. Apsidal rotation flips sign at a special inclination, coming up next. Both rates scale with (R_E / p)^2, so they are strongest for low, near-circular orbits and fade quickly with altitude.

Figure 2 plots both rates against inclination for a 700 km orbit. Two inclinations stand out, and each defines a famous orbit family.

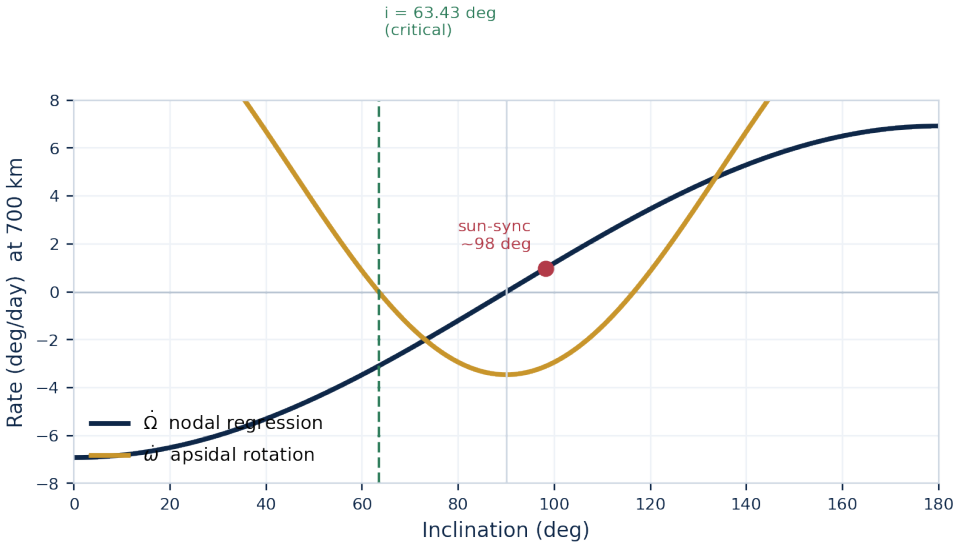



*Figure 2. Secular J2 rates versus inclination at 700 km. Nodal regression (navy) follows cos i: negative for prograde, zero at 90 degrees, positive for retrograde. Apsidal rotation (gold) follows 5cos squared i minus 1 and changes sign at the critical inclination.*

PART 3

# Turning Drift Into Design

## Sun-synchronous orbits

If we make the node drift eastward at exactly the rate the Earth orbits the Sun, the orbital plane keeps a constant angle to the Sun all year. The satellite then crosses every latitude at the same local solar time, giving consistent lighting: ideal for imaging and climate monitoring. The condition is a specific, small, positive nodal drift:

$$\dot{\Omega} = \frac{360^\circ}{365.25\ \text{d}} \approx +0.9856^\circ/\text{day}$$

Because a positive drift needs cos i < 0, sun-synchronous orbits are slightly retrograde and near-polar. Setting the J2 nodal-regression formula equal to that target gives, for a near-circular 700 to 800 km orbit, an inclination of about 98 degrees. That is why almost every Earth-observation satellite you can name sits near 98 degrees inclination.

## The critical inclination and Molniya orbits

The apsidal-rotation rate vanishes when its bracket is zero:

$$5\cos^2 i - 1 = 0 \Rightarrow i = 63.43^\circ \text{ or } 116.57^\circ$$

At the critical inclination the line of apsides stops rotating, so periapsis (and apoapsis) stay put. The Molniya orbit exploits this: a highly eccentric 12-hour orbit at i = 63.4 degrees, with

apoapsis frozen high over the northern hemisphere. The satellite loiters near apoapsis for hours of coverage at high latitudes that a GEO satellite cannot reach. This is the payoff of everything in Part 2: the perturbation you might have treated as a nuisance becomes the entire reason the orbit works.

**Worked estimate (ISS-like orbit)**
For a = 6,798 km, e ~ 0, i = 51.6 degrees, the secular J2 rates come out to RAAN drift of about -4.95 deg/day (a full turn of the node roughly every 73 days) and apsidal rotation of about +3.70 deg/day. You will reproduce these numbers, and confirm them by full numerical propagation, in this week code task.

PART 4

# Drag, Third-Body and Solar Radiation

## Atmospheric drag

At low altitude the thin upper atmosphere opposes the motion. Drag acceleration is:

$$\mathbf{a}_{\text{drag}} = -\frac{1}{2}\,\rho\,\frac{C_D A}{m}\,v_{\text{rel}}^2\,\hat{\mathbf{v}}_{\text{rel}}$$

where rho is the local air density and the ballistic coefficient bundles the drag coefficient, cross-sectional area and mass:

$$B = \frac{C_D A}{m}$$

Unlike J2, drag is not conservative: it removes energy, so the semi-major axis and eccentricity shrink, the orbit circularises and its altitude decays until re-entry. The effect is fiercest at perigee, where the satellite is lowest and fastest, and it depends strongly on solar activity, which heats and expands the atmosphere. This is why the ISS needs periodic reboosts and why orbit lifetime predictions carry large error bars.

## Third-body gravity (Sun and Moon)

For high orbits the Sun and Moon are no longer negligible. Their pull, relative to the Earth, is a tidal (difference) acceleration:

$$\mathbf{a}_{3b} = \mu_3\left(\frac{\mathbf{r}_3 - \mathbf{r}}{\|\mathbf{r}_3 - \mathbf{r}\|^3} - \frac{\mathbf{r}_3}{\|\mathbf{r}_3\|^3}\right)$$

Luni-solar attraction mainly changes inclination and RAAN of high orbits. A GEO satellite left alone drifts in inclination by roughly 0.75 to 0.95 degrees per year, which is the single largest station-keeping cost at GEO.

## Solar radiation pressure

Sunlight carries momentum. The push on a spacecraft depends on its reflectivity and, crucially, its area-to-mass ratio:

$$\mathbf{a}_{\text{SRP}} = -\,P_\odot\,C_R\,\frac{A}{m}\,\hat{\mathbf{s}}$$

SRP is tiny for compact satellites but dominates for large, light structures such as solar sails and big antennas, where it drives eccentricity oscillations that must be managed.

PART 5

## Station-Keeping and the Delta-v Budget

Station-keeping is the set of small manoeuvres that hold a satellite inside an allowed box against all of the above. Each manoeuvre costs a change in velocity, delta-v, and the mission carries a finite propellant budget, so the total is planned years in advance:

$$\Delta v_{\text{total}} = \sum_{k} |\Delta v_k|$$

The classic case is a geostationary satellite, which must be held both in latitude and longitude:

| Task | Fights | Typical cost |
|---|---|---|
| **North-South keeping** | Luni-solar inclination drift | about 45 to 55 m/s per year |
| **East-West keeping** | Longitude drift (Earth triaxiality) and SRP | about 2 to 4 m/s per year |
| **Total (GEO)** | All of the above | roughly 50 m/s per year |

North-South keeping dominates the budget, which is why designers sometimes allow the inclination to drift within a wider dead-band to save fuel. In LEO the equivalent job is drag make-up: periodic reboosts that restore the energy drag removes. Every one of these numbers is a vis-viva calculation from Session 1: you compute the speed change needed to move between the current orbit and the target, and that is your delta-v.

> **The through-line**
> Perturbations move the elements; station-keeping spends delta-v to move them back; vis-viva tells you how much each correction costs. Session 3 takes the same delta-v idea and uses it deliberately, to change orbits on purpose with Hohmann, bi-elliptic and low-thrust transfers. The GTO you sized last week is exactly a transfer orbit, so keep those numbers handy.

### Computing Perturbations

You can see J2 drift directly by propagating the full perturbed equation and reading the elements back out. The snippet below integrates two-body plus J2 from scratch with SciPy, so it does not depend on any one library version, then measures the nodal drift. Figure 3 is its output.

```
import numpy as np
from scipy.integrate import solve_ivp
mu, R_E, J2 = 398600.4418, 6378.137, 1.08262668e-3

def a_j2(r):
    x, y, z = r; rn = np.linalg.norm(r)
    c = -1.5 * J2 * mu * R_E**2 / rn**5
    return np.array([c*x*(1-5*z**2/rn**2),
                     c*y*(1-5*z**2/rn**2),
                     c*z*(3-5*z**2/rn**2)])

def rhs(t, s):
    r, v = s[:3], s[3:]; rn = np.linalg.norm(r)
    return np.hstack([v, -mu*r/rn**3 + a_j2(r)])

# propagate 3 days from an ISS-like state vector r0, v0
sol = solve_ivp(rhs, [0, 3*86400], np.hstack([r0, v0]),
                rtol=1e-10, atol=1e-10, max_step=60)
# recover RAAN from h = r x v at each step and compare to the secular rate
```

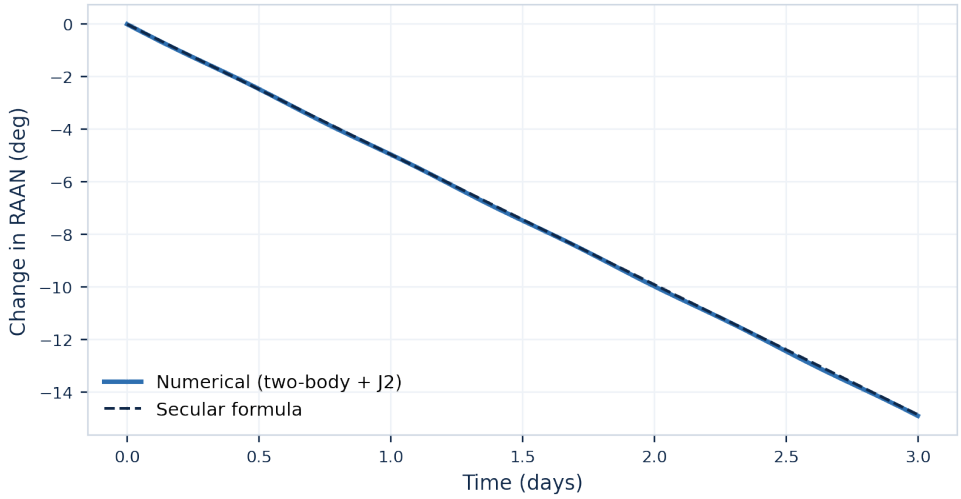



*Figure 3. RAAN from the full numerical propagation (blue) sits almost exactly on the secular formula (dashed) for the ISS-like orbit: about -4.95 deg/day. The tiny ripple around the line is the short-period J2 oscillation that the secular average smooths out.*

**Production tools**

In poliastro you would use the Cowell propagator with the built-in J2_perturbation function (the maintained fork hapsira has the same API). In STK, switch the propagator from TwoBody to J2, J4 or the High-Precision Orbit Propagator (HPOP) and enable the force models you need. GMAT exposes the same choices through its ForceModel resource. All three agree with the hand formulas to the precision of the force model you select.

PRACTICE

# This Week (aim for 12 to 15 hours)

Bring your work and plots to the next live session. As always, show your reasoning, not just the final numbers.

**Assignment: Perturbations and Station-Keeping**

- **Reading and notes (3 to 4 h).** In your own words, explain why J2 changes RAAN and argument of periapsis but not a, e or i on average. One page.
- **Secular rates by hand (3 to 4 h).** For the ISS-like orbit (a = 6,798 km, e = 0.001, i = 51.6 deg) compute the RAAN and argument-of-periapsis drift in deg/day. Then find the sun-synchronous inclination for a 700 km circular orbit and confirm the critical inclination of 63.4 deg from the formula.
- **Coding (4 to 5 h).** Complete the SciPy propagator above: recover RAAN from the state at each step, plot its drift over 3 days, and overlay the secular prediction. Your numerical slope should match to within a few percent. Then rerun at i = 98 deg and check the node now drifts the other way.
- **STK or GMAT (2 h).** Take your Session 1 ISS orbit, switch the propagator from TwoBody to J2 (or HPOP), run 7 days, and watch RAAN precess. Capture the before and after for your log.

## Answer key (for your self-check)

| Quantity | Expected value |
|---|---|
| **ISS-like RAAN drift** | about -4.95 deg/day (node circulates in ~73 days) |
| **ISS-like apsidal rotation** | about +3.70 deg/day |
| **Sun-sync inclination at 700 km** | about 98.2 deg |
| **Critical inclination** | 63.43 deg (and 116.57 deg) |
| **Numerical vs secular RAAN** | agree to about 0.3 percent over 3 days |

REFERENCE

# One-Page Formula Sheet

| | |
|---|---|
| **Perturbed equation of motion** | $\ddot{\mathbf{r}} = -\frac{\mu}{r^3}\mathbf{r} + \mathbf{a}_p$ |
| **Total perturbing acceleration** | $\mathbf{a}_p = \mathbf{a}_{J_2} + \mathbf{a}_{\text{drag}} + \mathbf{a}_{3-\text{body}} + \mathbf{a}_{\text{SRP}} + \cdots$ |
| **Nodal regression (RAAN)** | $\dot{\Omega} = -\frac{3}{2} J_2 \, n \left(\frac{R_E}{p}\right)^2 \cos i$ |
| **Apsidal rotation** | $\dot{\omega} = \frac{3}{4} J_2 \, n \left(\frac{R_E}{p}\right)^2 (5\cos^2 i - 1)$ |
| **Mean motion, semi-latus rectum** | $n = \sqrt{\frac{\mu}{a^3}}, \qquad p = a(1-e^2)$ |
| **Critical inclination** | $5\cos^2 i - 1 = 0 \Rightarrow i = 63.43^\circ$ or $116.57^\circ$ |
| **Sun-synchronous condition** | $\dot{\Omega} = \frac{360^\circ}{365.25 \text{ d}} \approx +0.9856^\circ/\text{day}$ |
| **Atmospheric drag** | $\mathbf{a}_{\text{drag}} = -\frac{1}{2} \rho \frac{C_D A}{m} v_{\text{rel}}^2 \hat{\mathbf{v}}_{\text{rel}}$ |
| **Third-body (tidal)** | $\mathbf{a}_{3b} = \mu_3 \left( \frac{\mathbf{r}_3 - \mathbf{r}}{\lVert \mathbf{r}_3 - \mathbf{r} \rVert^3} - \frac{\mathbf{r}_3}{\lVert \mathbf{r}_3 \rVert^3} \right)$ |
| **Solar radiation pressure** | $\mathbf{a}_{\text{SRP}} = -P_\odot C_R \frac{A}{m} \hat{\mathbf{s}}$ |

## Key glossary

- **Zonal harmonic J2:** the leading term describing Earth equatorial bulge; J2 = 1.0826 x 10^-3.
- **Secular vs periodic:** secular effects grow without bound over time; periodic effects oscillate and average to zero.
- **Nodal regression:** the steady drift of RAAN caused by J2.
- **Ballistic coefficient:** C_D A / m; a high value means drag matters more.
- **Dead-band:** the allowed box around the target slot before a station-keeping burn is triggered.

## Going further

For depth: Curtis, Orbital Mechanics for Engineering Students (Chapter on orbital perturbations); Vallado, Fundamentals of Astrodynamics and Applications (perturbation methods and force models); and Montenbruck and Gill, Satellite Orbits, for the numerical side. These are real, widely used references, but please confirm exact editions and chapters yourself, as I cannot verify those details from here.

**Coming up in Session 3**

Orbit transfers and delta-v: Hohmann and bi-elliptic transfers, plane changes, and an introduction to low-thrust. We will reuse vis-viva heavily and start from the very GTO you sized in Session 1, turning it into a full Earth-to-GEO transfer with a real delta-v cost.